

The de Branges index of the continuation

Hermite–Biehler gates hold; phase dominance with index ≤ 1 is refuted

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Abstract

After rounds 413/415 closed the hole-top-mode route, the reviewer (DCCLXXIX, path B) asks for the last unused structural language: de Branges / Potapov on the two finite positive OP systems of the continuation ($X = \mu$ -chain on the X -nodes with dual weights $u^\vee$; $Y = \text{hole measure}$). The candidate lemma has three equivalent forms: a contractive embedding $H(E_Y) \hookrightarrow H(E_X)$ up to one mode; a rational transfer matrix Θ with Potapov index $\kappa(\Theta) \leq 1$; Prüfer phase dominance $\varphi'_X \geq \varphi'_Y$ except one deficit of total drop $< 2\pi$. The index is basis-free (no exceptional vector). The Hermite–Biehler pair of each positive discrete measure is a finite SATZ (zeros of p_n and q_n real and interlacing; Wronskian constant over $\mathbb{Q}$). The phase-dominance lemma is *refuted* on flagship MAIN: the combinatorial Rücklauf counter (YY-adjacencies of $\text{Gauss}_X(d_0)$ against the hole atoms) equals 3, not ≤ 1 , and is uncorrelated with $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0)$. Permute does not inflate the counter. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Propositions 1–3 are finite machine checks. Ausgang PHASE_DOMINANCE_REFUTED / HB_CENSUS. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Definitions

Let Y be the hole nodes of a window, X the complementary dual nodes, and $u^\vee$ the r356/r407 dual weights. Write $q = |Y|$ and $d_0 = N_w - 3$ for the dual interpolating depth of $\mathfrak{T}_0$ (r409). The X -system is the Jacobi chain of $(X, u^\vee|_X)$ at degree $n_X = \min(d_0, |X| - 1)$; the Y -system is the chain of $(Y, u^\vee|_Y)$ at degree $n_Y = q - 1$. First-kind orthonormal polynomials p_n and second-kind q_n (Cauchy transform against p) give the Hermite–Biehler function $E = A - iB$ with $A = p_n$, $B = q_n$. Gauss zeros of p_n are the eigenvalues of the $n \times n$ Jacobi matrix; zeros of q_n are those of the trailing principal submatrix. The combinatorial Rücklauf counter yy_A is the number of consecutive Y -atoms with no X -Gauss node between them in the merged ordered list (one mode $\Leftrightarrow \text{yy}_A \leq 1$). The equal-dimension control yy_S merges $\text{Gauss}_X(q - 1)$ with $\text{Gauss}_Y(q - 1)$.

2 Finite identities

Lemma 1 (Wronskian). *On the r409 rational 3-atom X -measure, the monic pair $p_2(z) = (z - a_1)(z - a_0) - \gamma_1$, $q_2(z) = z - a_1$ has Wronskian $p_2q_1 - p_1q_2 = -\gamma_1$ independent of z , with $\gamma_1 = 252000/3463321 > 0$.*

Lemma 2 (interlacing over $\mathbb{Q}$). *In the same toy, $\text{disc}(p_2) = (a_0 - a_1)^2 + 4\gamma_1 = 17761/19600 > 0$, and $p_2(a_1) = -\gamma_1 < 0$, so the unique zero of q_2 strictly interlaces the two zeros of p_2 .*

Lemma 3 (Hermite–Biehler of a positive Jacobi matrix). *For every measured positive discrete measure in the census (MAIN 42, living χ 78, dead χ 6, permute, scramble, flagship w_9 , the 5-atom toy) the zeros of p_n and q_n are real and interlace. This is the classical Gauss / Gauss–Radau relation for a positive Jacobi matrix, gated rather than assumed.*

Remark 1. Lemma 3 is automatic for any positive measure. It does *not* constrain the relative zero sets of the X -system and the Y -system, which is where the index lemma lives.

3 Machine checks

Proposition 1 (degree balance). *At $k_z = 9$, $|X| = 263$, $|Y| = 104$, $d_0 = 181$, so $n_X = d_0 = 181 < |X| - 1$ (the X -chain is truncated to the interpolating depth) and $n_Y = 103 = q - 1$. Both Stieltjes transforms are Herglotz at $z = 0.3 + 0.7i$ ($\text{Im } m_X, \text{Im } m_Y > 0$), while $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) = 1$ and $\|\mathfrak{T}_0\| = 1.08014$. The full-measure Uvarov path (r392 elementary Stieltjes steps) has Potapov index 0 and cannot see the truncated interpolant.*

Proposition 2 (primary test). *On flagship MAIN, $yy_A = 3 \not\leq 1$ and $yy_S = 24 \gg 1$. The phase counter is not the operator index ($yy_A \neq \text{ind}_-(I - \mathfrak{T}^* \mathfrak{T})$). The dominant YY -interval versus the top right singular vector of $\mathfrak{T}_0$ has relative location 0.092 at $k_z = 9$, but across MAIN the location spans $[0.059, 1]$: it is not a coordinate that walks with the window-individual defect.*

Proposition 3 (census and kills). *MAIN 42: Hermite–Biehler 42/42; yy_A in $\{0, \dots, 15\}$ with median 4; of 28 P1 windows only 3 have $yy_A \leq 1$; of 14 PD windows only 3 have $yy_A = 0$; $\text{corr}(yy_A, \text{ind}_-(I - \mathfrak{T}^* \mathfrak{T})) = -0.17$. Living χ (78): yy_A median 7, max 26. Dead χ (6): median 2, max 4 — death is fewer Rückläufe, not a bulk-phase overflow. Permute: $yy_A = 4$ (the MAIN median) with $\text{ind}_- = 20$ — the world-separator does not fire on the phase counter. Scramble: $yy_A = 16 \neq 21$. The mutant $B := p_n$ (no second kind) breaks interlacing by degree.*

4 What is a theorem, what is a census

The counting-argument candidate — interlacing of X -Gauss zeros against Y -atoms as a corollary of the fold partition of the ordered node set — is closed: the fold does not force $yy_A \leq 1$. Hermite–Biehler of each system separately is a theorem (Lemmas 1–3). The YY table is a census. The Potapov index of $\mathfrak{T}_0$ itself remains $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0)$ (r409 dictionary); re-reading it as a de Branges phase of the two OP systems does not produce a new P1 proof. P1 stays $\lambda_2(C) \geq 1$ on the dual quotient. No RH claim.